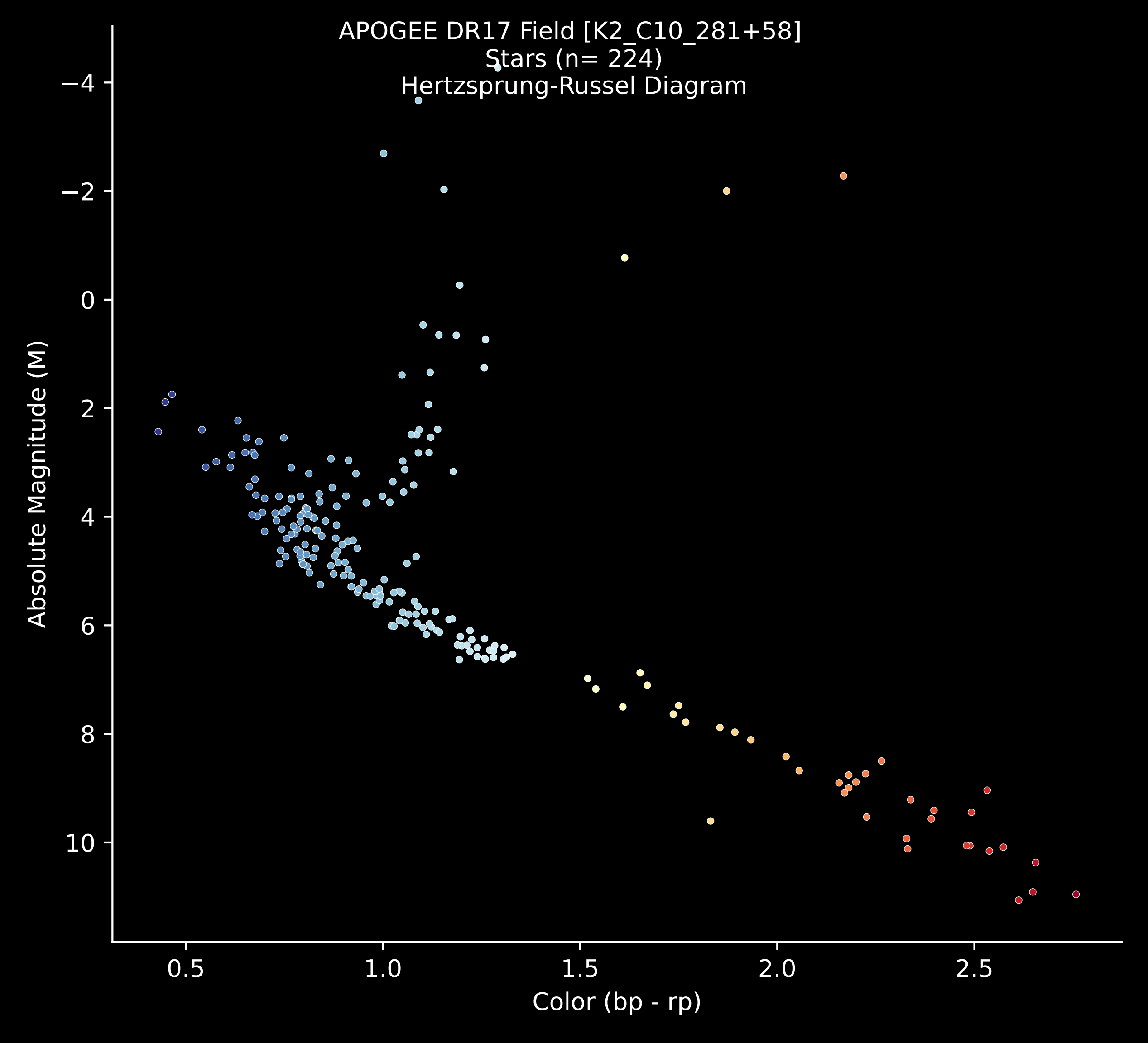
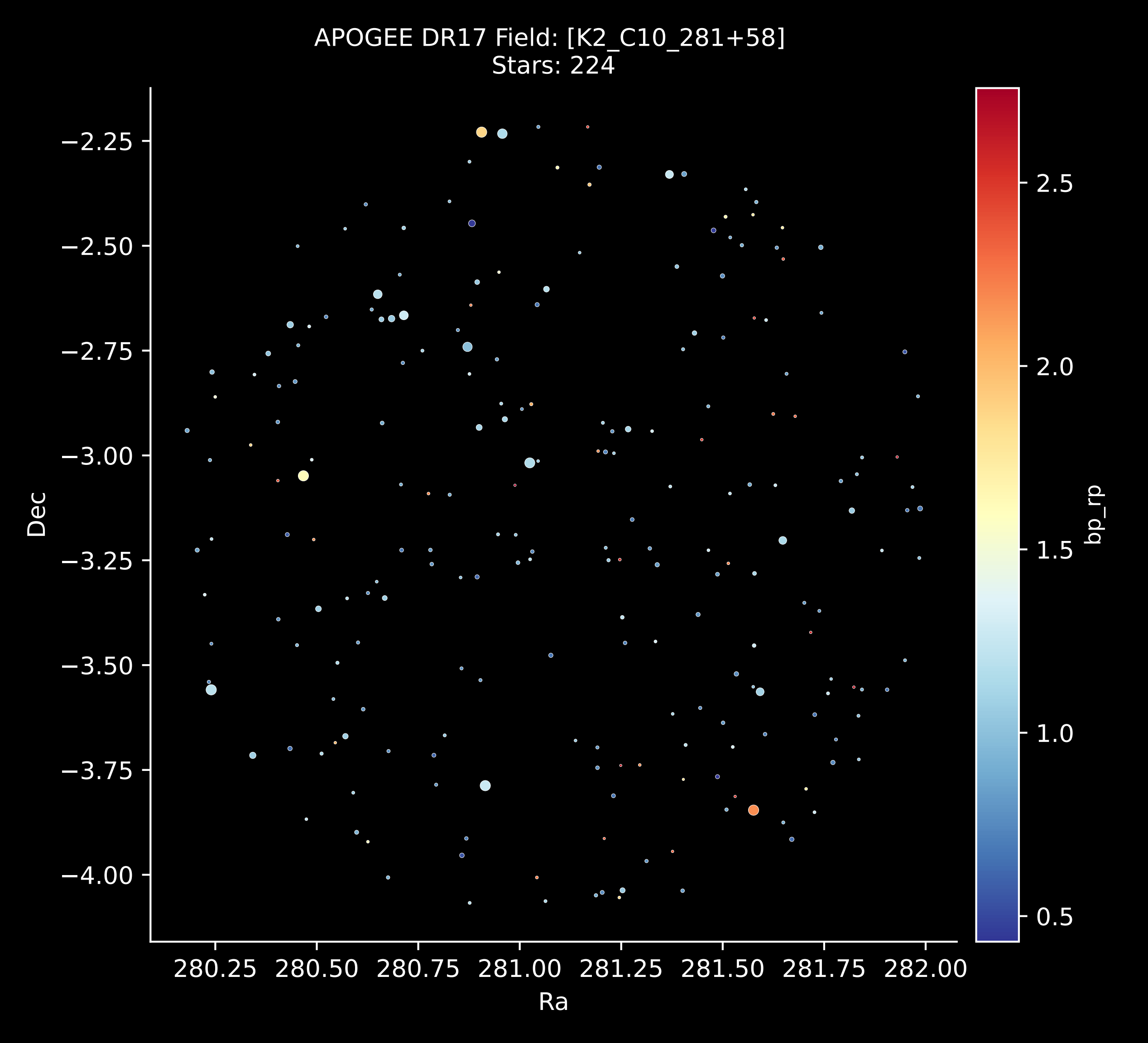
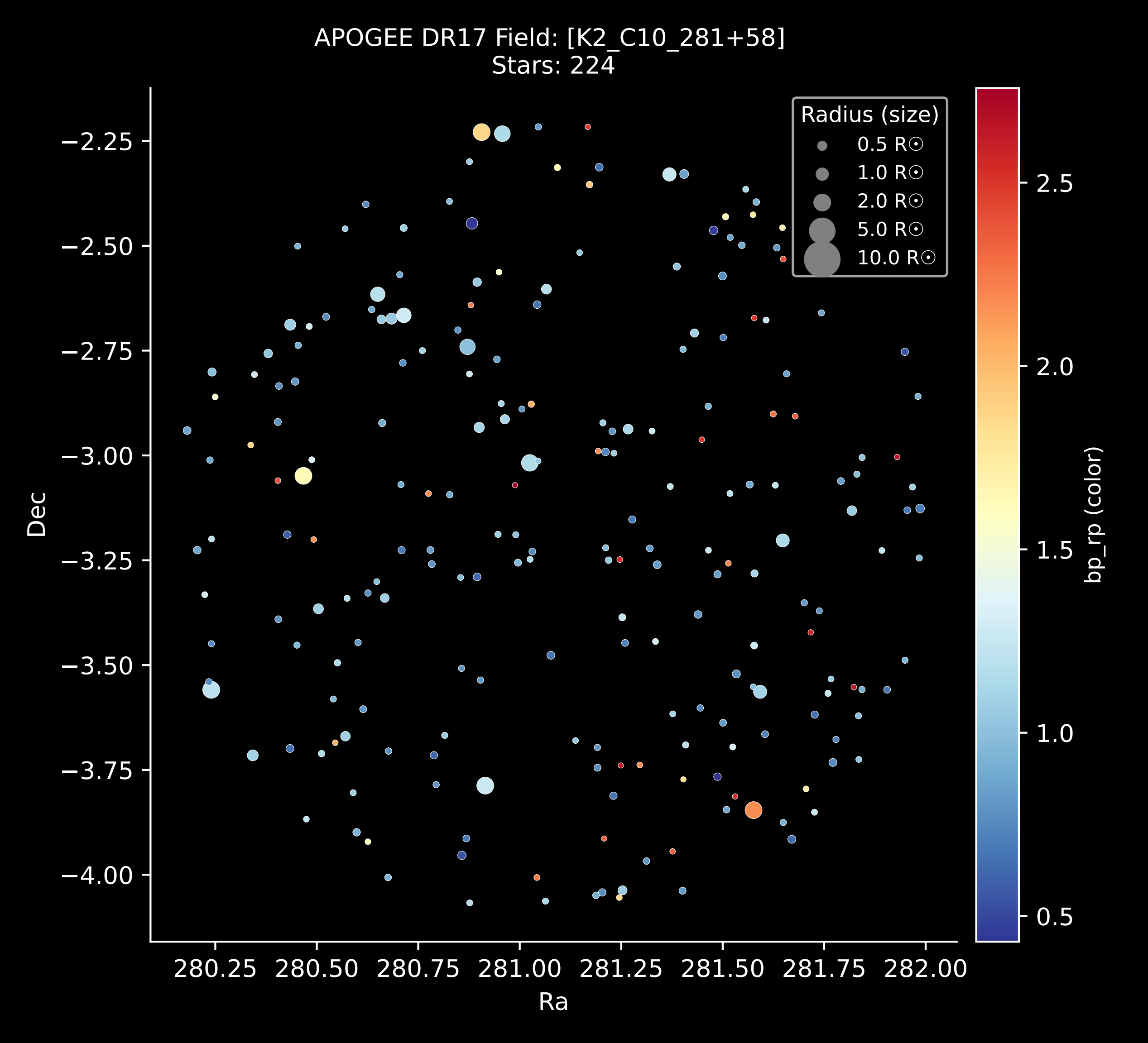
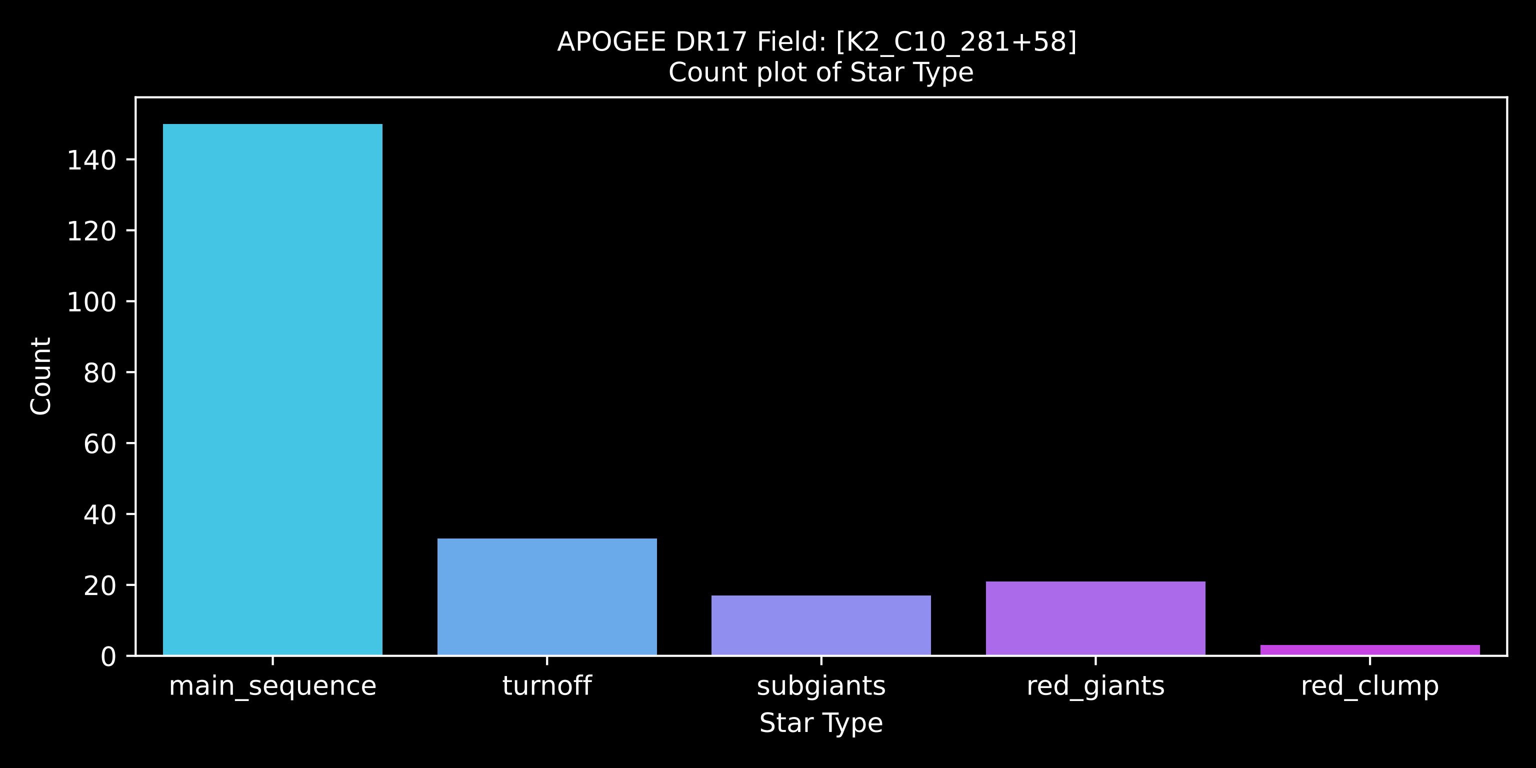
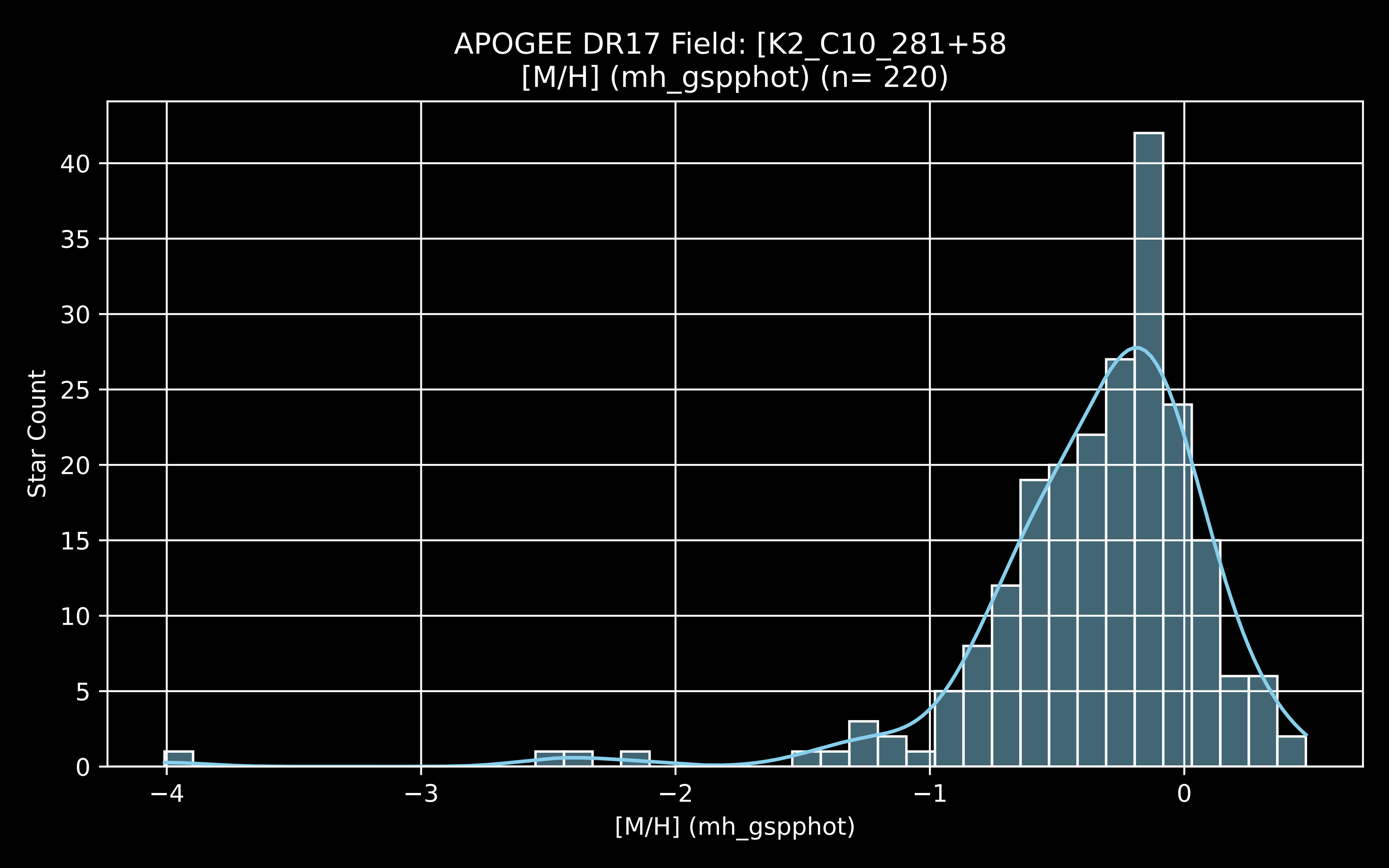

APOGEE DR17 Field [K2_C10_281+58]
Stars (n= 224)
Hertzsprung-Russel Diagram
Absolute Magnitude (M)
Color (bp - rp)
−4
−2
0
2
4
6
8
10
0.5
1.0
1.5
2.0
2.5


APOGEE DR17 Field: [K2_C10_281+58]
Stars: 224
Dec
Ra
bp_rp
−2.25
−2.50
−2.75
−3.00
−3.25
−3.50
−3.75
−4.00
280.25
280.50
280.75
281.00
281.25
281.50
281.75
282.00
2.5
2.0
1.5
1.0
0.5

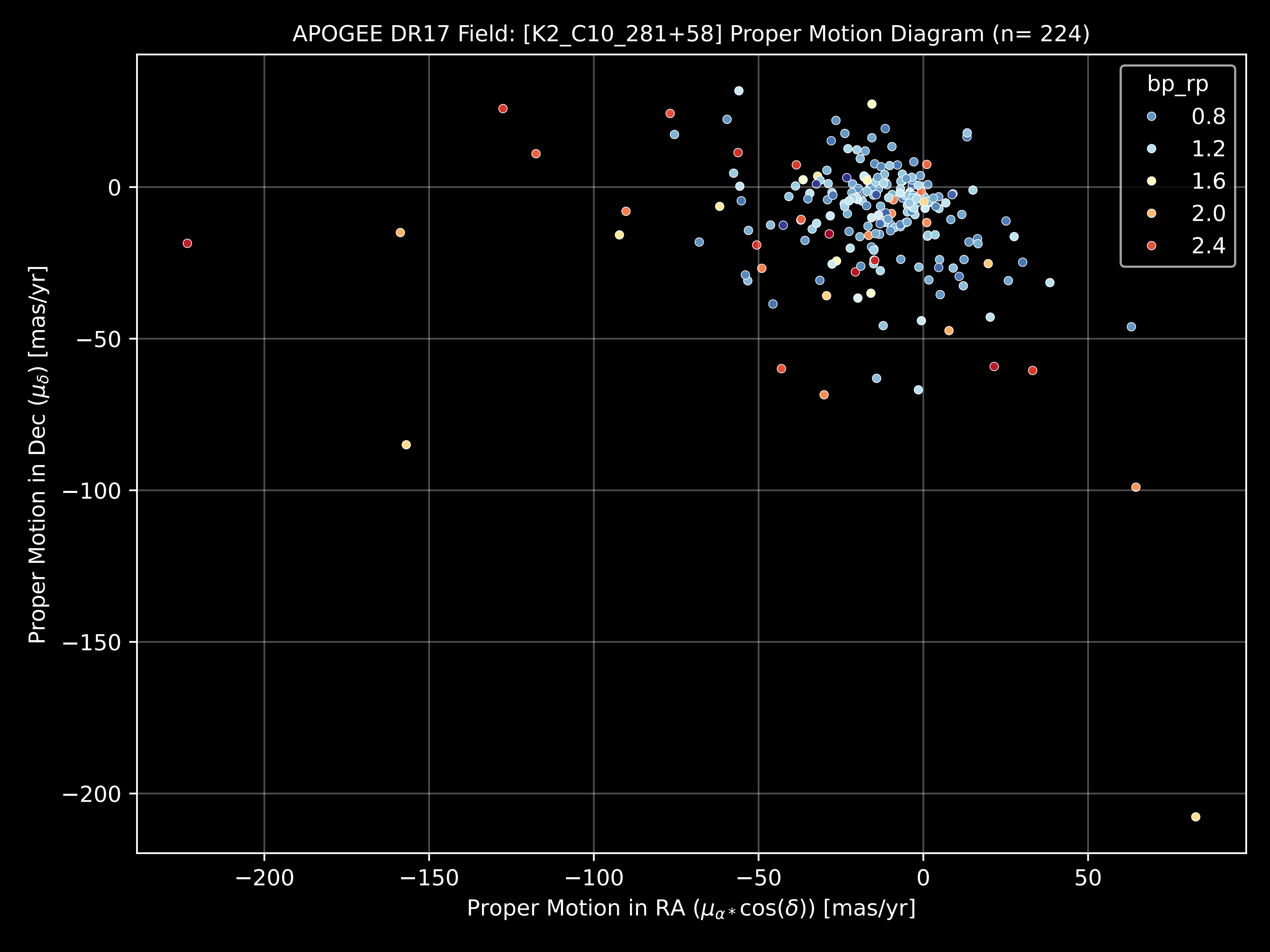


APOGEE DR17 Field: [K2_C10_281+58]
Stars: 224
Radius (size)
0.5 R☉
1.0 R☉
2.0 R☉
5.0 R☉
10.0 R☉
Dec
Ra
bp_rp (color)
−2.25
−2.50
−2.75
−3.00
−3.25
−3.50
−3.75
−4.00
280.25
280.50
280.75
281.00
281.25
281.50
281.75
282.00
0.5
1.0
1.5
2.0
2.5

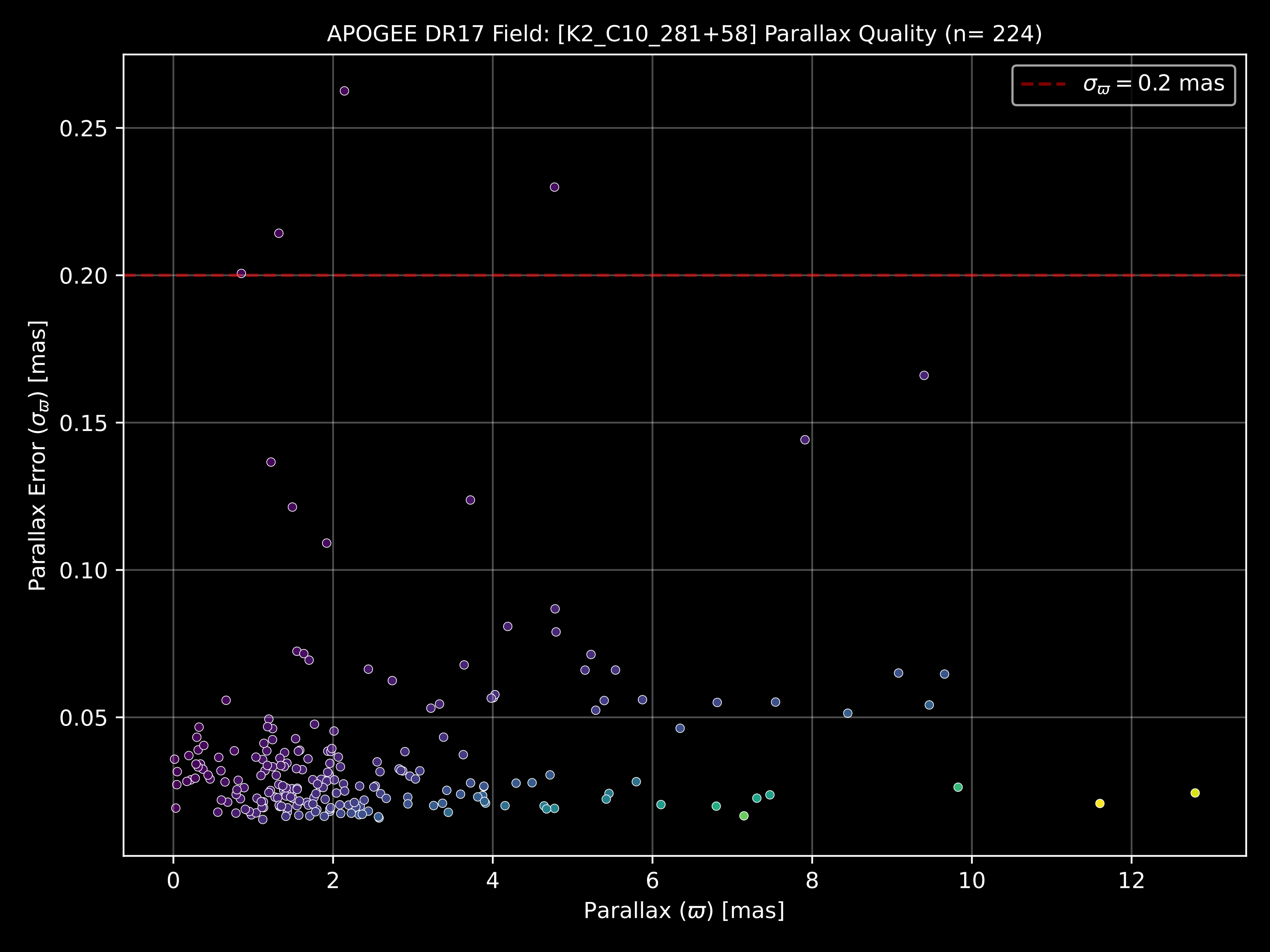

APOGEE DR17 Field: [K2_C10_281+58]
Count plot of Star Type
Count
0
20
40
60
80
100
120
140
main_sequence
turnoff
subgiants
red_giants
red_clump
Star Type



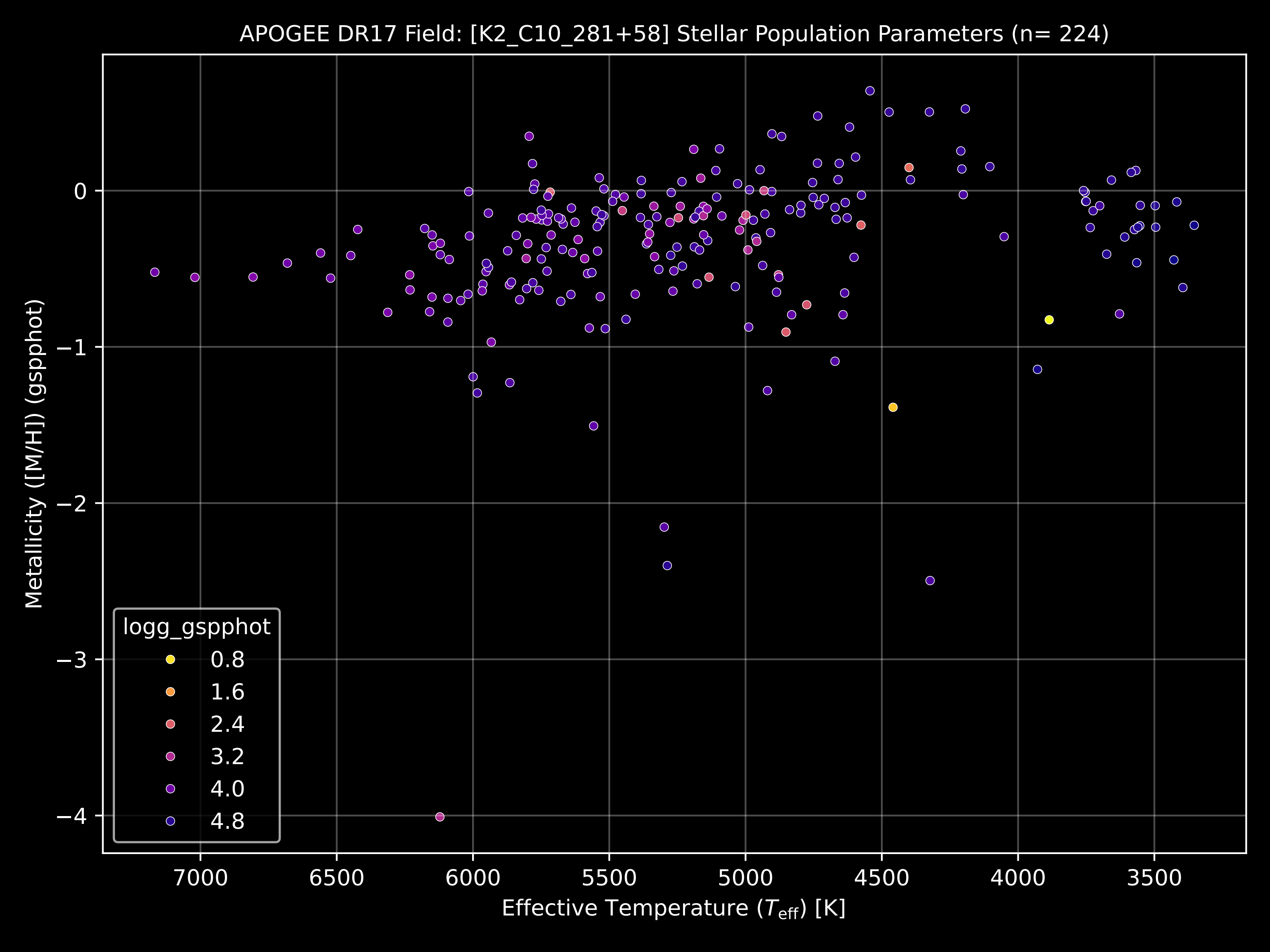
APOGEE DR17 Field: [K2_C10_281+58
[M/H] (mh_gspphot) (n= 220)
Star Count
[M/H] (mh_gspphot)
0
5
10
15
20
25
30
35
40
−4
−3
−2
−1
0



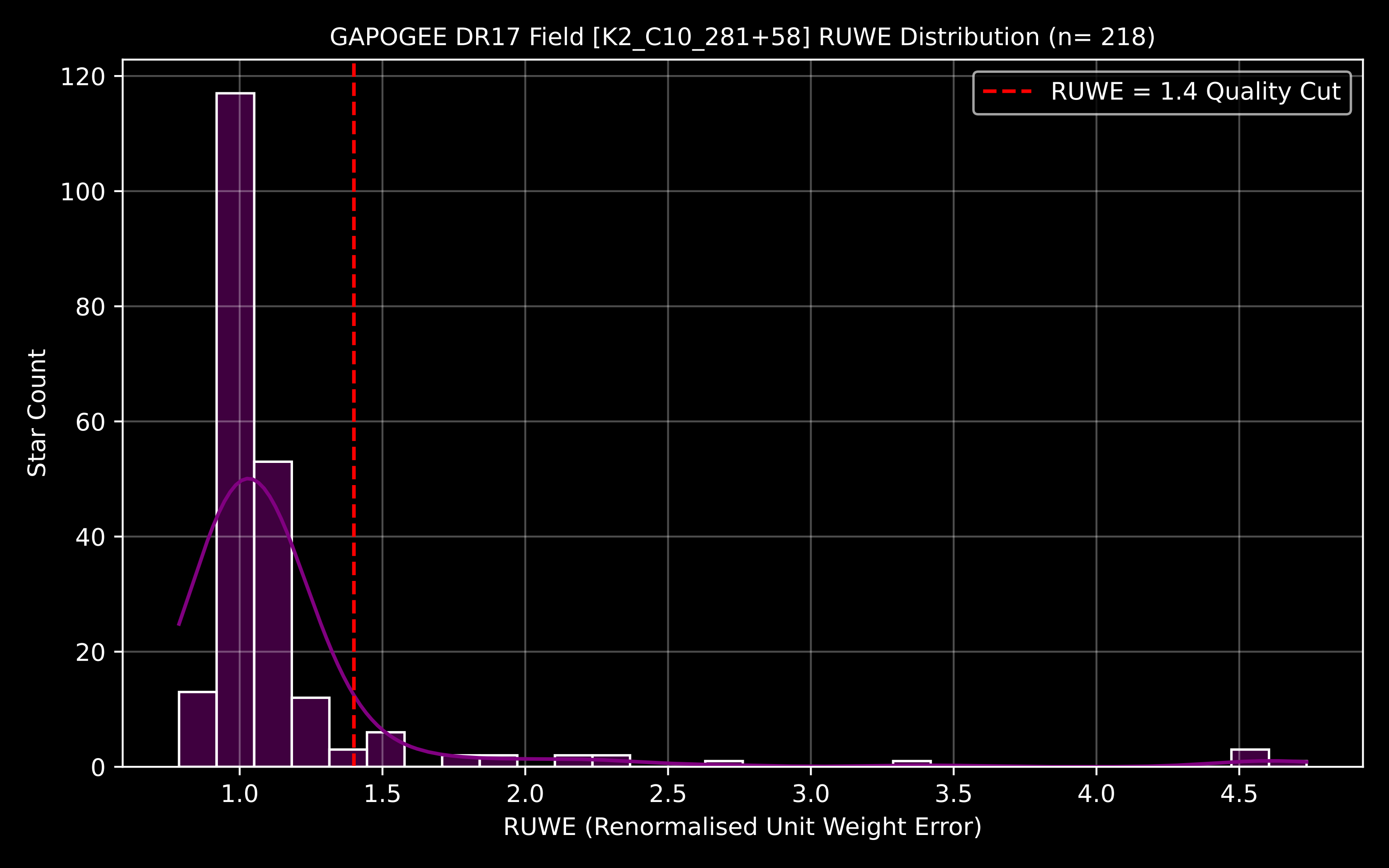
APOGEE DR17 Field: [K2_C10_281+58] Proper Motion Diagram (n= 224)
Proper Motion in Dec ($\mu_\delta$) [mas/yr]
Proper Motion in RA ($\mu_{\alpha*}\cos(\delta)$) [mas/yr]
bp_rp
0.8
1.2
1.6
2.0
2.4
0
−50
−100
−150
−200
50

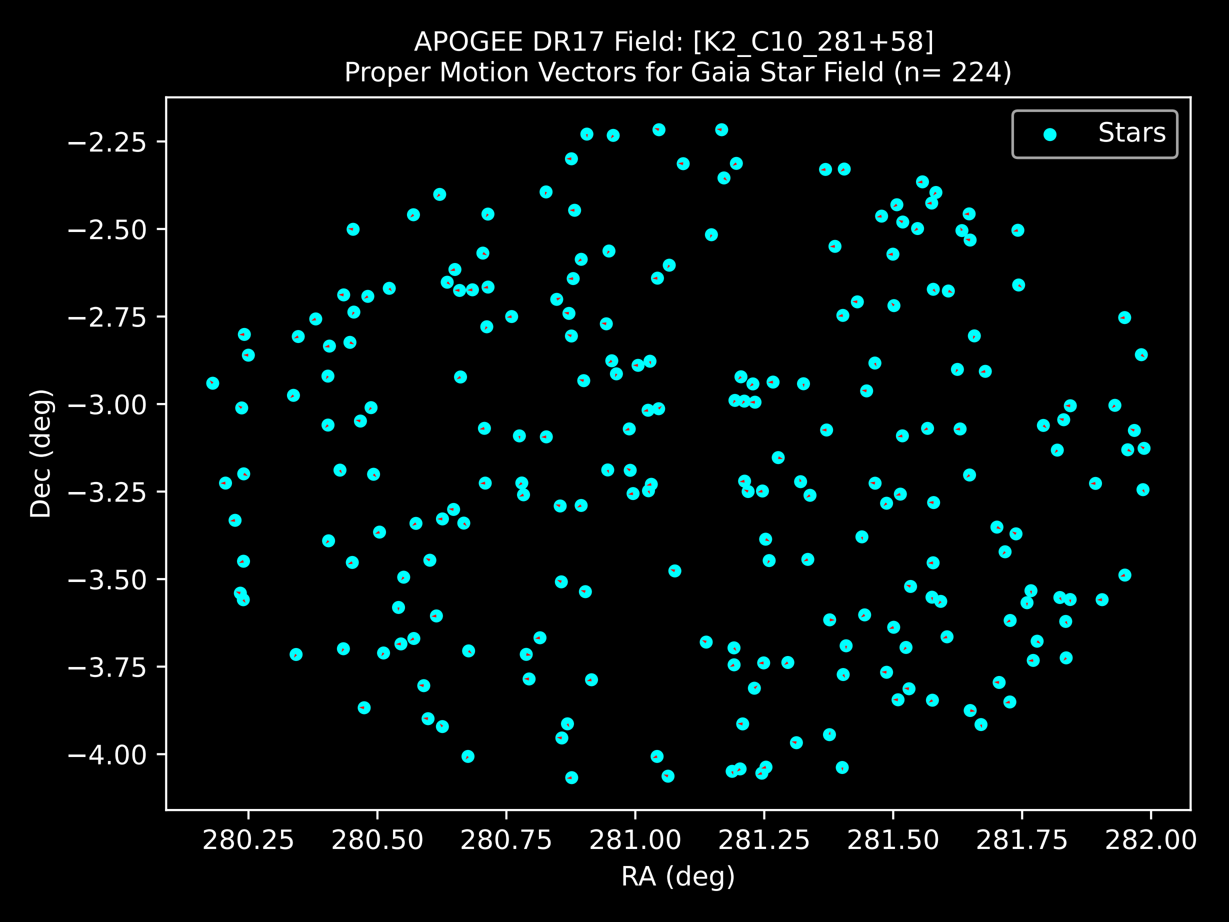


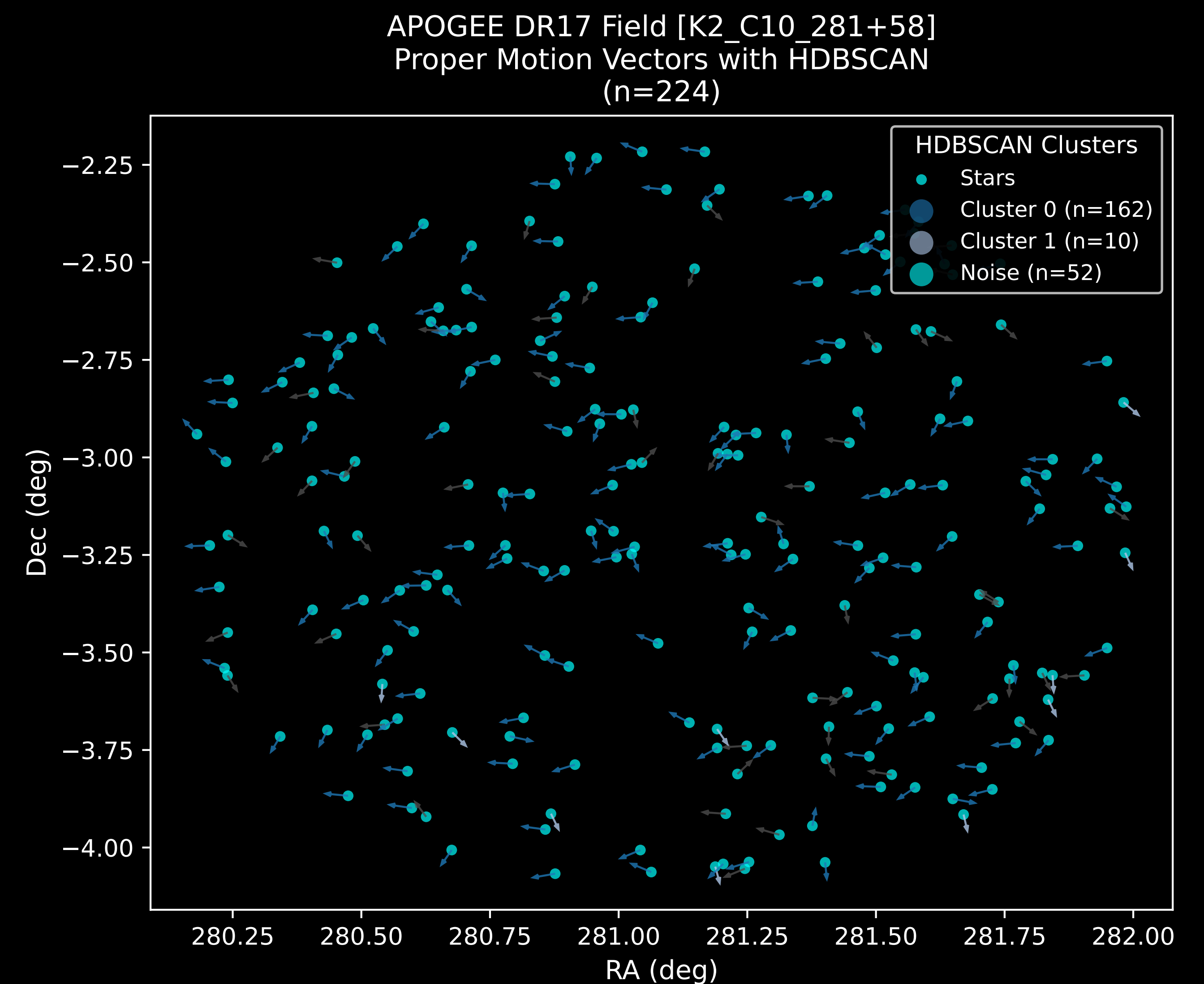
APOGEE DR17 Field: [K2_C10_281+58] Parallax Quality (n= 224)
$\sigma_\varpi = 0.2$ mas
Parallax Error ($\sigma_\varpi$) [mas]
0.25
0.20
0.15
0.10
0.05
0
2
4
6
8
10
12
Parallax ($\varpi$) [mas]

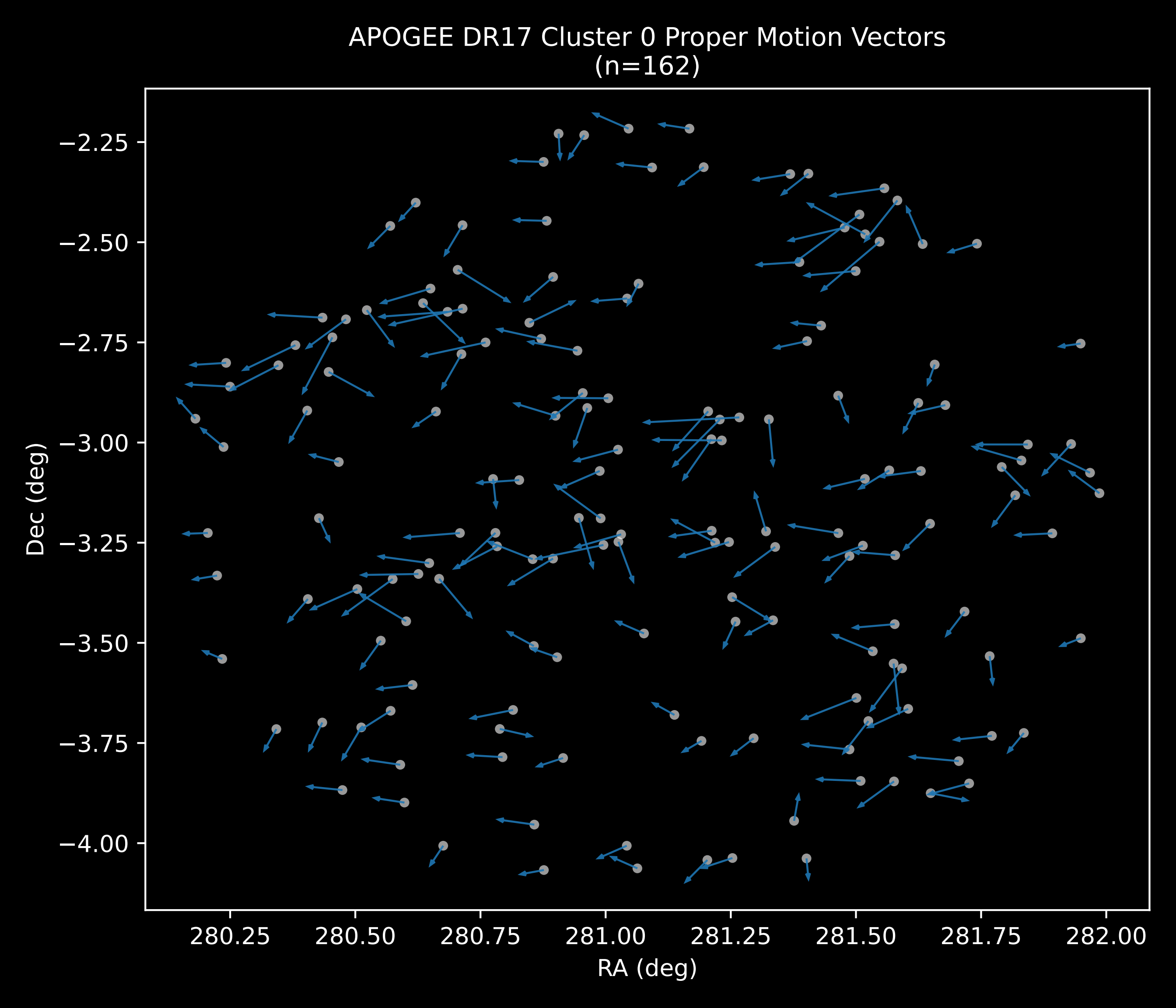


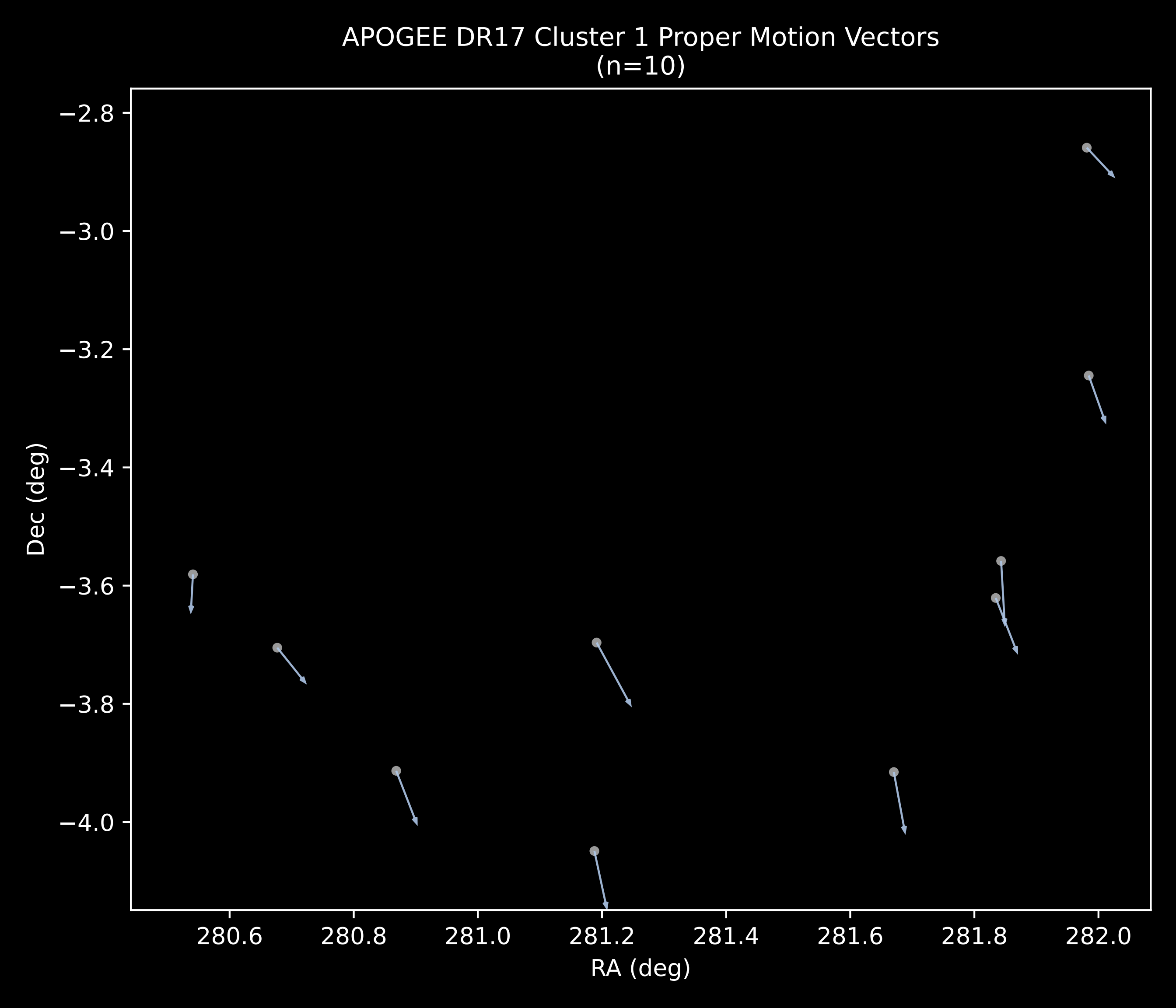
APOGEE DR17 Field: [K2_C10_281+58] Stellar Population Parameters (n= 224)
Metallicity ([M/H]) (gspphot)
Effective Temperature ($T_{\mathrm{eff}}$) [K]
0
−1
−2
−3
−4
7000
6500
6000
5500
5000
4500
4000
3500
logg_gspphot
0.8
1.6
2.4
3.2
4.0
4.8


GAPOGEE DR17 Field [K2_C10_281+58] RUWE Distribution (n= 218)
RUWE = 1.4 Quality Cut
Star Count
RUWE (Renormalised Unit Weight Error)
0
20
40
60
80
100
120
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5

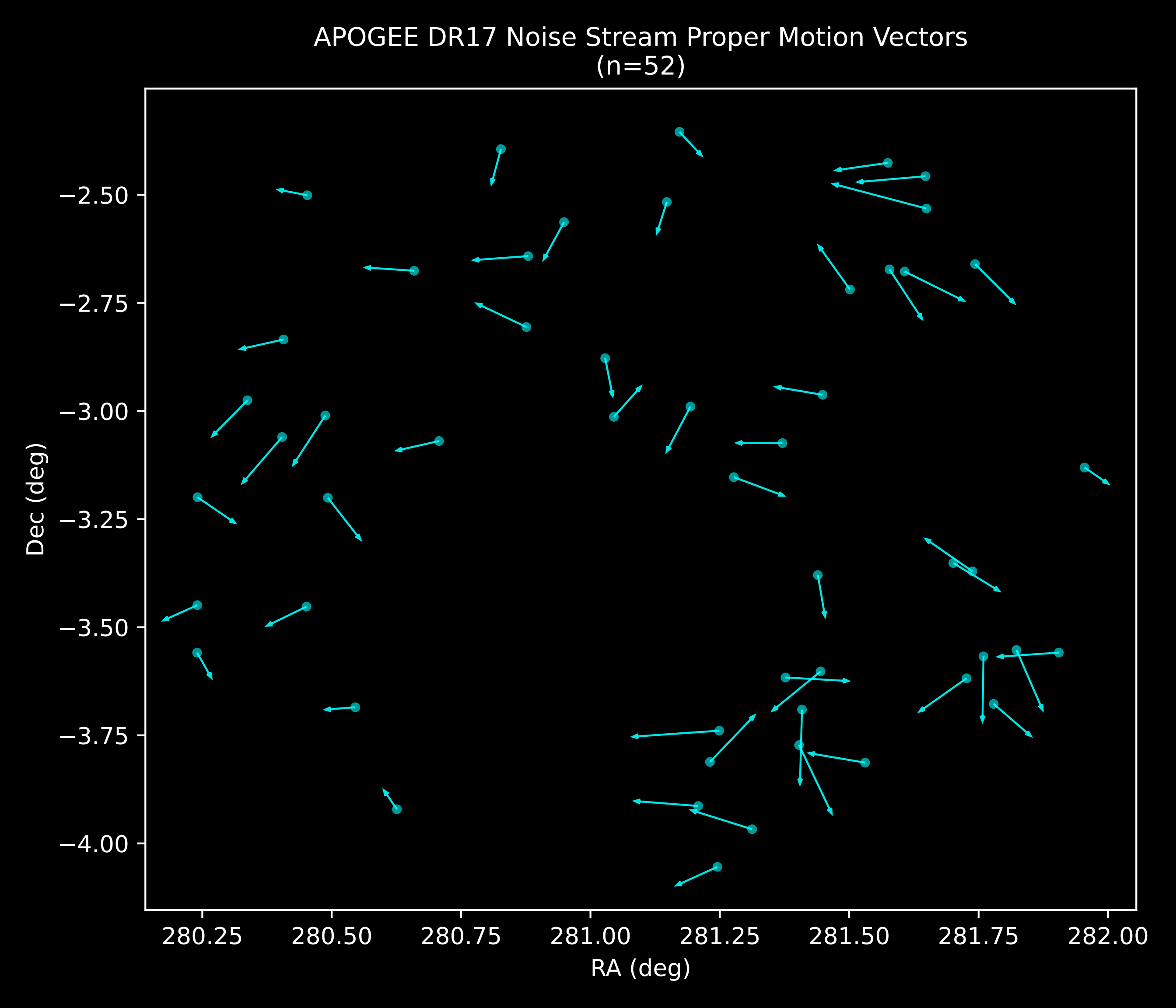

APOGEE DR17 Field: [K2_C10_281+58]
Proper Motion Vectors for Gaia Star Field (n= 224)
Stars
Dec (deg)
RA (deg)
−2.25
−2.50
−2.75
−3.00
−3.25
−3.50
−3.75
−4.00
280.25
280.50
280.75
281.00
281.25
281.50
281.75
282.00


APOGEE DR17 Field [K2_C10_281+58]
Proper Motion Vectors with HDBSCAN
(n=224)
HDBSCAN Clusters
Stars
Cluster 0 (n=162)
Cluster 1 (n=10)
Noise (n=52)
Dec (deg)
RA (deg)
−2.25
−2.50
−2.75
−3.00
−3.25
−3.50
−3.75
−4.00
280.25
280.50
280.75
281.00
281.25
281.50
281.75
282.00


APOGEE DR17 Cluster 0 Proper Motion Vectors
(n=162)
Dec (deg)
RA (deg)
−2.25
−2.50
−2.75
−3.00
−3.25
−3.50
−3.75
−4.00
280.25
280.50
280.75
281.00
281.25
281.50
281.75
282.00



APOGEE DR17 Cluster 1 Proper Motion Vectors
(n=10)
−2.8
−3.0
−3.2
−3.4
−3.6
−3.8
−4.0
Dec (deg)
280.6
280.8
281.0
281.2
281.4
281.6
281.8
282.0
RA (deg)


APOGEE DR17 Noise Stream Proper Motion Vectors
(n=52)
Dec (deg)
−2.50
−2.75
−3.00
−3.25
−3.50
−3.75
−4.00
RA (deg)
280.25
280.50
280.75
281.00
281.25
281.50
281.75
282.00